

Nonlinear Control of a Space Robot with Slosh Dynamics in Two Dimension

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Abstract—Slosh dynamics could make some harmful changes in attitude parameters of space robots in orbital maneuvering. So, it is necessary to control this harmful phenomenon. This paper presents a mathematical modeling and nonlinear control of sloshing and attitude coupled dynamics of a space robot in two dimension. Fuzzy and Lyapunov controllers are utilized for attitude stabilizing of the space robot. Effectiveness of this modeling and controllers are illustrated by the simulation results.

Index Terms—fluid dynamics, fuzzy controller, Lyapunov controller, attitude control

I. INTRODUCTION

Today with development of space missions many attractive fields such as collision in space [1]–[3], spacecraft formation [4]–[6], site location in LEO missions [7]–[10] and orbital maneuvering [11], [12] are introduced. Attitude control is the most important problem in every space mission [13]–[17], especially in orbital maneuvering [18], [19]. Usually, attitude control is done by designing a control law for using thrusters [20], magnetorquers [21] and reaction wheels [15], whereas some of these laws are difficult to design [22]–[26]. Also, attitude control has wide range of usage in aerial objects [27]–[29]. As for orbital maneuvering, the space robot which maneuvers in space needs liquid fuel to introduce thrust force. This fuel consists higher percentage of space robot mass than other sub-systems, and could harmfully change attitude of the space robot, if axial or rotational acceleration of the space robot changes [30]. This problem could bring heavy recompense. Therefore, it is necessary to control slosh dynamics. Interaction between attitude and slosh dynamics is a problem that studied in many papers. Most of these studies proved that mechanical model like pendulum and mass-spring model, can approximate fluid slosh dynamics in orbital maneuvering.

Some of articles studied single pendulum [31], [32] or mass-spring [32] to describe first fluid sloshing mode, and other describe most prominent fluid sloshing modes using multi-pendulum [33]–[35] or mass-spring [36], [37].

In current study, a two-pendulum model in two dimension is introduced to describe two prominent modes of sloshing in a space robot. The space robot is maneuvering in 2D space. As seen in Fig. 1 the space robot is denoted by a rigid mass (m) that is located in center of space robot mass, and as seen in Fig. 2 fluid is denoted by two parts: un-turbulent and

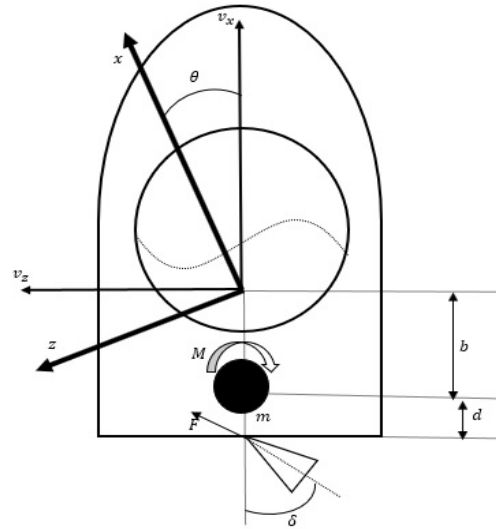


Fig. 1. Maneuvering space robot and its container

sloshing fluid, and they are demonstrated by rigidly attached mass (m_0) and pendulum masses (m_i), respectively. After modeling, fuzzy and Lyapunov controllers are designed to suppress attitude of the space robot and pendulums. In this paper, gimbal deflection angle (δ) and moment about center of the space robot mass (M) are control inputs. After designing controllers, effectiveness of them on attitude of the spacecraft and pendulums is illustrated by simulation.

II. TWO-PENDULUM MODEL

Equations of coupled dynamics of the space robot and fluid dynamics are obtained using means of Lagrange equations. The space robot is denoted by a rigid mass (m, I) which is located in center of the space robot mass, and fluid is denoted by a rigidly attached mass (m_0, I_0) which demonstrates un-turbulent fuel, and two pendulum masses ($m_i, I_i, i = 1, 2$) which are demonstrate sloshing fuel. Equations of coupled dynamics are derived by Kirchoffs equations and extension of Lagrangian to include non-conservative forces.

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{V}} + \bar{\Omega} \times \frac{\partial L}{\partial \dot{V}} = \tau_t \quad (1)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\Omega}} + \bar{\Omega} \times \frac{\partial L}{\partial \dot{\Omega}} + \bar{V} \times \frac{\partial L}{\partial \dot{V}} = \tau_r \quad (2)$$

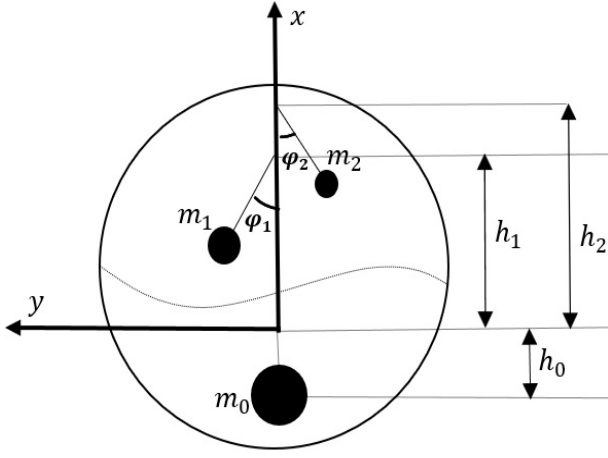


Fig. 2. Schematic of two-pendulum model describing modes of sloshing

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\zeta}} - \frac{\partial L}{\partial \zeta} + \frac{\partial R}{\partial \dot{\zeta}} = 0 \quad (3)$$

Where $V = [v_x \ 0 \ v_z]^T$, $\Omega = [0 \ \Omega \ 0]^T$ are vectors of axial and angular velocity of the rigid mass, respectively. $\zeta = [\varphi_1 \ \varphi_2]^T$ is vector of pendulum mass attitudes, and τ_t, τ_r are force vector and moment vector acting on rigid mass. Also, $\bar{\Omega}$ and $\bar{V}$ are skew-symmetric matrices. L is Lagrangian and R is Rayleigh dissipation function, and they are written respectively:

$$L = \frac{1}{2}m(v_x^2 + (v_z + b\Omega)^2) + \frac{1}{2}m_0(v_x^2 + (v_z + h_0\Omega)^2) + \frac{1}{2}(I + I_0)\Omega^2 + \frac{1}{2} \sum_{i=1}^2 [m_i((v_x + l_i(\Omega + \dot{\varphi}_i) \sin \varphi_i)^2 + (v_z - h_i\Omega + l_i(\Omega + \dot{\varphi}_i) \cos \varphi_i)^2) + I_i(\Omega + \dot{\varphi}_i)^2] \quad (4)$$

$$R = \sum_{i=1}^2 \epsilon_i \dot{\varphi}_i^2 \quad (5)$$

Where h_0, h_i are distances between fuel masses and center of container, and l_i is length of pendulum masses. Also, ϵ_i is damping coefficient of pendulum masses. By applying (1-3) the equations of coupled dynamics are derived as:

$$(m + m_0)a_x + mb\Omega^2 + \sum_{i=1}^2 [m_i a_x + m_i l_i (\dot{\Omega} + \ddot{\varphi}_i) \sin \varphi_i - m_i l_i \Omega^2 + m_i l_i (\Omega + \dot{\varphi}_i)^2 \cos \varphi_i] = F \cos \delta \quad (6)$$

$$(m + m_0)a_z + mb\dot{\Omega} + \sum_{i=1}^2 [m_i l_i (\dot{\Omega} + \ddot{\varphi}_i) \cos \varphi_i - m_i l_i \dot{\Omega} - m_i l_i (\Omega + \dot{\varphi}_i)^2 \sin \varphi_i] = F \sin \delta \quad (7)$$

$$(mb^2 + m_0 h_0^2) \dot{\Omega} + (I + I_0) \dot{\Omega} + mba_z - \sum_{i=1}^2 [m_i h_i^2 \dot{\Omega} + m_i l_i h_i ((\dot{\Omega} + \ddot{\varphi}_i) \cos \varphi_i - (\Omega + \dot{\varphi}_i)^2 \sin \varphi_i) - m_i l_i a_z - \epsilon_i \dot{\varphi}_i] = M + F(b + d) \sin \delta \quad (8)$$

$$(I_i + m_i l_i^2)(\dot{\Omega} + \ddot{\varphi}_i) - m_i l_i h_i (\dot{\Omega} \cos \varphi_i + \Omega^2 \sin \varphi_i) + m_i l_i (a_x \sin \varphi_i + a_z \cos \varphi_i) + \epsilon_i \dot{\varphi}_i = 0 \quad (9)$$

Where b is distance between center of container and center of the space robot mass, and d is distance between center of the space robot mass and gimbal pivot. Also, $a_x = \dot{v}_x + \Omega v_z$ and $a_z = \dot{v}_z - \Omega v_x$ are accelerations of coordinates origin.

III. CONTROLLER DESIGNING

There are two designed controllers to suppress attitude of the space robot and pendulums. Due to maneuvering in space it is not necessary to suppress v_x . So, the first equation of equations of coupled dynamics is ignored. Now by changing [31] the control inputs to new control inputs $(u_{1,2})$, equations of motion are rewritten as following.

$$\dot{v}_z = u_1 + v_x \Omega \quad (10)$$

$$\dot{\Omega} = u_2 \quad (11)$$

$$\ddot{\varphi}_i = -\frac{m_i l_i}{I_i + m_i l_i^2} u_1 \cos \varphi_i - \frac{a_x m_i l_i}{I_i + m_i l_i^2} \sin \varphi_i - u_2 + \frac{m_i l_i h_i}{I_i + m_i l_i^2} (\cos \varphi_i u_2 + \Omega^2 \sin \varphi_i) - \frac{\epsilon_i}{I_i + m_i l_i^2} \dot{\varphi}_i \quad (12)$$

A. Fuzzy Controller

In this sub-section fuzzy controller is designed to achieve control aims. Therefore, new control inputs are written by P and PD control theory.

$$u_1 = k_{p1} v_z \quad (13)$$

$$u_2 = k_{p2} \theta + k_{d2} \Omega \quad (14)$$

$$k_{p1} = k_{p1min} + (k_{p1max} - k_{p1min}) k'_{p1} \quad (15)$$

$$k_{p2} = k_{p2min} + (k_{p2max} - k_{p2min}) k'_{p2} \quad (16)$$

$$k_{d2} = k_{d2min} + (k_{d2max} - k_{d2min}) k'_{d2} \quad (17)$$

$$k_{p1min} = 0.64 k_{p1}, k_{p1max} = 1.2 k_{p1}$$

$$k_{p2min} = 0.53 k_{p2}, k_{p2max} = k_{p2}$$

$$k_{d2min} = k_{d2}, k_{d2max} = 2 k_{d2}$$

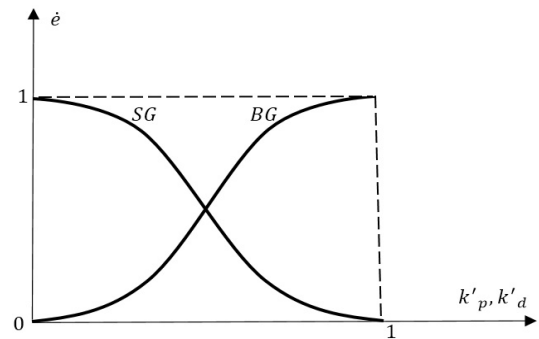


Fig. 3. Gaussian membership functions

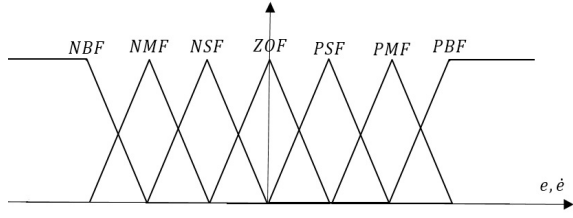


Fig. 4. Five triangle and two trapeze membership functions

Where θ is attitude angle of the space robot. k'_{pi} , k'_{d2} are output gains of Mamdani logic, and as shown in Fig. 3 they have two Gaussian membership functions. Inputs of this logic are v_z , θ , and Ω , and as seen in Fig. 4, they have five triangle and two trapeze membership functions.

The rules are defined using Tab. I and Tab. II [38]. Also, for defining the rules between k'_{p1} and v_z , if v_z was in biggest membership (NBF or PBF), k'_{p1} is BG, else k'_{p1} is SG.

TABLE I
RULES FOR k'_p

$\dot{e}$		NBF	NMF	NSF	ZOF	PSF	PMF	PBF
e	NBF	BG	BG	BG	BG	BG	BG	BG
	NMF	SG	BG	BG	BG	BG	BG	SG
	NSF	SG	SG	BG	BG	BG	SG	SG
	ZOF	SG	SG	SG	BG	SG	SG	SG
	PSF	SG	SG	BG	BG	BG	SG	SG
	PMF	SG	BG	BG	BG	BG	BG	SG
	PBF	BG	BG	BG	BG	BG	BG	BG

TABLE II
RULES FOR k'_d

$\dot{e}$		NBF	NMF	NSF	ZOF	PSF	PMF	PBF
e	NBF	SG	SG	SG	SG	SG	SG	SG
	NMF	BG	BG	SG	SG	SG	BG	BG
	NSF	BG	BG	BG	SG	BG	BG	BG
	ZOF	BG	BG	BG	BG	BG	BG	BG
	PSF	BG	BG	BG	SG	BG	BG	BG
	PMF	BG	BG	SG	SG	SG	BG	BG
	PBF	SG	SG	SG	SG	SG	SG	SG

B. Lyapunov Controller

In this section Lyapunov controller is designed to achieve control aims. At first, a candidate Lyapunov function is introduced.

$$\mathbf{V} = \frac{1}{2}r_1v_z^2 + \frac{1}{2}r_2\theta^2 + \frac{1}{2}r_3\Omega^2 + \frac{1}{2}r_4\left(\sum_{i=1}^2[\dot{\varphi}_i^2 + \dot{\varphi}_i\Omega(1 - \frac{m_i h_i l_i}{I_i + m_i l_i^2} \cos \varphi_i)]\right) \quad (18)$$

Time differential of candidate function is derived as

$$\dot{\mathbf{V}} = [L] + [M]u_1 + \sum_{i=1}^2[(f u_2 + g)\Omega] \quad (19)$$

Where $[L]$, $[M]$, f and g are functions of the system variables. Then the new control inputs are written as below

$$u_1 = -w_1[M] \quad (20)$$

$$u_2 = -\frac{1}{f}(g + w_2\Omega) \quad (21)$$

IV. SIMULATION

Now for illustrating effectiveness of designed controllers, a simulation is conducted. Here, the purposes of designed controller are suppressing attitude of the space robot and pendulums. The control gains of fuzzy and Lyapunov controllers are given in Tab. III and Tab. IV, respectively. Also, physical parameters are given in Tab. V.

TABLE III
FUZZY CONTROLLER GAINS

$k_{p1}=6 \times 10^{-3}$	
$k_{p2}=5 \times 10^{-5}$	$k_{d2}=6 \times 10^{-3}$

TABLE IV
LYAPUNOV CONTROLLER GAINS

$r_1=1.25 \times 10^{-6}$	$r_4=1 \times 10^{-3}$
$r_2=400$	$w_1=6 \times 10^3$
$r_3=500$	$w_2=1 \times 10^4$

TABLE V
PHYSICAL PARAMETERS

m	590 Kg	h_0	0.05 m
I	400 Kg.m ²	h_1	0.6 m
m_0	480 Kg	h_2	0.9 m
I_0	75 Kg.m ²	F	2250 N
m_1	50 Kg	l_1	0.2 m
I_1	10 Kg.m ²	l_2	0.1 m
m_2	5 Kg	ϵ_1	3.7 Kg.m ² /s
I_2	1 Kg.m ²	ϵ_2	0.5 Kg.m ² /s

So, by applying designed controllers on reduced form equations, time variations of the space robot and pendulums variables will be shown. As seen in Fig. 5, v_z and θ considering fuzzy and Lyapunov controllers trend to zero after elapse 1200 and 700 seconds, and it is clear that Lyapunov controller has better performance than fuzzy. Moreover, as indicated in Fig. 6, the attitude angle of the pendulums are suppressed after elapse the same time, also Lyapunov controller has better performance. Furthermore, time variations of control inputs (M and δ) are shown in Fig. 7.

V. CONCLUSION

A complete nonlinear mathematical model of coupled dynamics of the space robot and fluid sloshing has been developed by two-pendulum model in 2D space, and fuzzy and Lyapunov controllers have been designed to control attitude of the space robot and pendulums. The effectiveness of designed controllers has been illustrated by a simulation, and its shown that designed controllers have good responses at control aims.

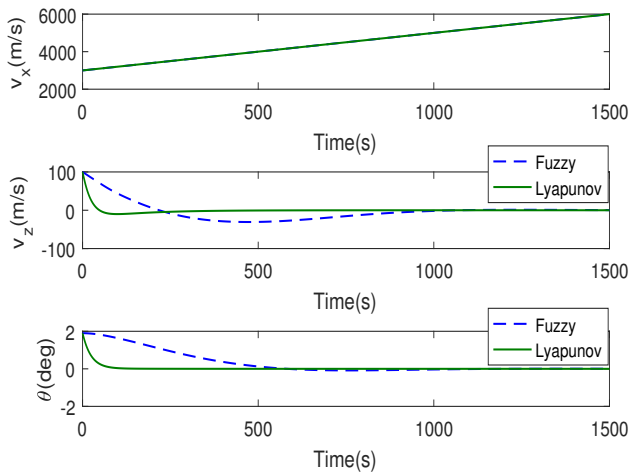


Fig. 5. Time variations for v_z and θ

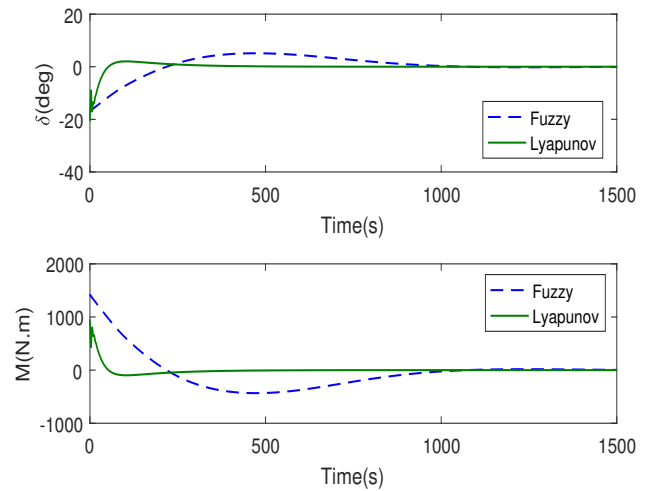


Fig. 7. Time variations for M and δ

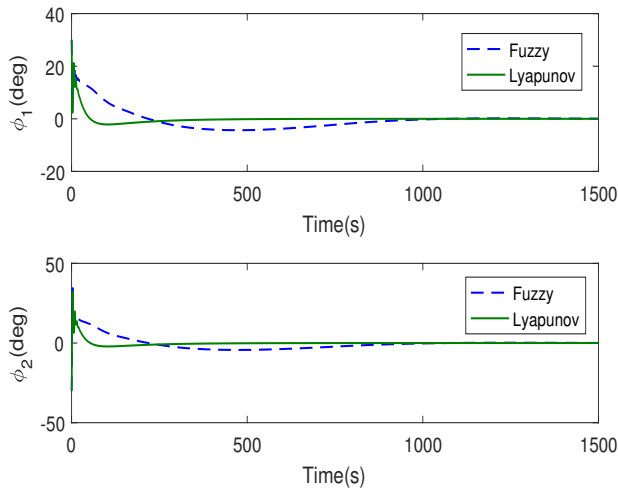


Fig. 6. Time variations for attitude angle of the pendulums (ϕ_i)

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